

Chi-Square Goodness of Fit Testing in Mendelian Genetics and Hardy-Weinberg Equilibrium

Contents

1	Part 1: Testing Mendelian Crosses	3
1.1	Example 1: Monohybrid Cross (Complete Dominance)	3
1.2	Example 2: Monohybrid Cross (Co-dominance)	3
1.3	Example 3: Dihybrid Cross	3
1.4	Example 4: Test Cross (1:1 Ratio)	4
2	Part 2: Testing Hardy-Weinberg Equilibrium	5
2.1	Example 5: Testing HWE at a Codominant Locus	5
2.2	Example 6: Testing HWE with a Heterozygote Deficiency	5
2.3	Example 7: Testing HWE with a Rare Recessive Trait	6
3	Part 3: Degrees of Freedom in HWE Testing - A Detailed Discussion	7
3.1	General Principle	7
3.2	Why This Matters	7
3.3	Example 8: Allele Frequency Estimated from Data ($df = 1$)	7
3.4	Example 9: Allele Frequency Known from External Source ($df = 2$)	8
3.5	Example 10: Multiple Alleles with Known Frequencies	8
3.6	Summary Table	9
4	Practice Problems	10
5	Answer Key	12

Introduction

The chi-square (χ^2) goodness of fit test is used to determine whether observed categorical data differ significantly from expected values under a specific hypothesis. In genetics, we commonly use this test in two contexts:

1. Testing Mendelian inheritance patterns (monohybrid, dihybrid, codominance)
2. Testing whether a population is in Hardy-Weinberg equilibrium

The formula for the chi-square statistic is:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

where:

- O = observed count for each category
- E = expected count for each category under the null hypothesis
- The sum is taken over all categories

Degrees of Freedom

- For Mendelian crosses: $df = (\text{number of phenotypic classes}) - 1$
- For Hardy-Weinberg with two alleles: df depends on how expected values are derived (see Section 3 for details)

Decision Rule

If the calculated χ^2 value is greater than the critical value from the chi-square distribution table (typically at $\alpha = 0.05$), we reject the null hypothesis. Otherwise, we fail to reject the null hypothesis.

Table 1: Critical values of χ^2 at $\alpha = 0.05$

Degrees of Freedom	Critical Value
1	3.84
2	5.99
3	7.81
4	9.49
5	11.07

1 Part 1: Testing Mendelian Crosses

1.1 Example 1: Monohybrid Cross (Complete Dominance)

A geneticist crosses two heterozygous pea plants ($Aa \times Aa$) for flower color, where purple (A) is dominant over white (a). The expected phenotypic ratio is 3:1. In the offspring, she observes 96 purple flowers and 24 white flowers (total $N = 120$).

Step 1: State the null hypothesis H_0 : The data follow a 3:1 Mendelian ratio.

Step 2: Calculate expected counts Expected purple: $\frac{3}{4} \times 120 = 90$ Expected white: $\frac{1}{4} \times 120 = 30$

Step 3: Calculate chi-square

$$\chi^2 = \frac{(96 - 90)^2}{90} + \frac{(24 - 30)^2}{30} = \frac{36}{90} + \frac{36}{30} = 0.4 + 1.2 = 1.6$$

Step 4: Determine degrees of freedom $df = 2 - 1 = 1$

Step 5: Compare to critical value Critical value at $df = 1$ is 3.84. Since $1.6 < 3.84$, we **fail to reject** the null hypothesis. The data are consistent with a 3:1 Mendelian ratio.

1.2 Example 2: Monohybrid Cross (Co-dominance)

In cattle, coat color shows codominance at the roan locus: $C^R C^R$ = red, $C^R C^W$ = roan, $C^W C^W$ = white. A cross between two roan cattle ($C^R C^W \times C^R C^W$) is expected to produce a 1:2:1 ratio. A farmer observes 25 red, 48 roan, and 27 white calves (total $N = 100$).

Step 1: State the null hypothesis H_0 : The data follow a 1:2:1 ratio.

Step 2: Calculate expected counts Expected red: $\frac{1}{4} \times 100 = 25$ Expected roan: $\frac{2}{4} \times 100 = 50$ Expected white: $\frac{1}{4} \times 100 = 25$

Step 3: Calculate chi-square

$$\chi^2 = \frac{(25 - 25)^2}{25} + \frac{(48 - 50)^2}{50} + \frac{(27 - 25)^2}{25} = \frac{0}{25} + \frac{4}{50} + \frac{4}{25} = 0 + 0.08 + 0.16 = 0.24$$

Step 4: Determine degrees of freedom $df = 3 - 1 = 2$

Step 5: Compare to critical value Critical value at $df = 2$ is 5.99. Since $0.24 < 5.99$, we **fail to reject** the null hypothesis. The data are consistent with a 1:2:1 codominant ratio.

1.3 Example 3: Dihybrid Cross

In pea plants, a dihybrid cross ($AaBb \times AaBb$) for seed shape (round vs. wrinkled) and seed color (yellow vs. green) is expected to produce a 9:3:3:1 ratio. A researcher observes: 315 round yellow, 108 round green, 101 wrinkled yellow, 32 wrinkled green (total $N = 556$).

Step 1: State the null hypothesis H_0 : The data follow a 9:3:3:1 ratio.

Step 2: Calculate expected counts Expected round yellow: $\frac{9}{16} \times 556 = 312.75$
Expected round green: $\frac{3}{16} \times 556 = 104.25$ Expected wrinkled yellow: $\frac{3}{16} \times 556 = 104.25$
Expected wrinkled green: $\frac{1}{16} \times 556 = 34.75$

Step 3: Calculate chi-square

$$\begin{aligned}
 \chi^2 &= \frac{(315 - 312.75)^2}{312.75} + \frac{(108 - 104.25)^2}{104.25} \\
 &\quad + \frac{(101 - 104.25)^2}{104.25} + \frac{(32 - 34.75)^2}{34.75} \\
 &= \frac{(2.25)^2}{312.75} + \frac{(3.75)^2}{104.25} + \frac{(-3.25)^2}{104.25} + \frac{(-2.75)^2}{34.75} \\
 &= \frac{5.0625}{312.75} + \frac{14.0625}{104.25} + \frac{10.5625}{104.25} + \frac{7.5625}{34.75} \\
 &= 0.016 + 0.135 + 0.101 + 0.218 = 0.470
 \end{aligned}$$

Step 4: Determine degrees of freedom $df = 4 - 1 = 3$

Step 5: Compare to critical value Critical value at $df = 3$ is 7.81. Since $0.470 < 7.81$, we **fail to reject** the null hypothesis. The data are consistent with a 9:3:3:1 ratio.

1.4 Example 4: Test Cross (1:1 Ratio)

A test cross ($Aa \times aa$) is expected to produce a 1:1 phenotypic ratio. A breeder observes 142 dominant phenotype and 158 recessive phenotype (total $N = 300$).

Step 1: State the null hypothesis H_0 : The data follow a 1:1 ratio.

Step 2: Calculate expected counts Expected dominant: 150 Expected recessive: 150

Step 3: Calculate chi-square

$$\chi^2 = \frac{(142 - 150)^2}{150} + \frac{(158 - 150)^2}{150} = \frac{64}{150} + \frac{64}{150} = 0.427 + 0.427 = 0.854$$

Step 4: Determine degrees of freedom $df = 2 - 1 = 1$

Step 5: Compare to critical value Critical value at $df = 1$ is 3.84. Since $0.854 < 3.84$, we **fail to reject** the null hypothesis. The data are consistent with a 1:1 ratio.

2 Part 2: Testing Hardy-Weinberg Equilibrium

For a locus with two alleles, we test whether observed genotype frequencies match HWE expectations. Important: When estimating allele frequencies from the data, we lose one degree of freedom.

2.1 Example 5: Testing HWE at a Codominant Locus

In a population of 100 individuals, the following MN blood group genotypes are observed: MM = 36, MN = 48, NN = 16. Test whether this population is in Hardy-Weinberg equilibrium.

Step 1: Calculate observed allele frequencies Total alleles = $2 \times 100 = 200$
Frequency of M: $p = \frac{2 \times 36 + 48}{200} = \frac{72 + 48}{200} = \frac{120}{200} = 0.6$ Frequency of N: $q = 1 - 0.6 = 0.4$

Step 2: Calculate expected genotype counts under HWE Expected MM: $p^2 \times 100 = 0.36 \times 100 = 36$ Expected MN: $2pq \times 100 = 0.48 \times 100 = 48$ Expected NN: $q^2 \times 100 = 0.16 \times 100 = 16$

Step 3: Calculate chi-square

$$\chi^2 = \frac{(36 - 36)^2}{36} + \frac{(48 - 48)^2}{48} + \frac{(16 - 16)^2}{16} = 0 + 0 + 0 = 0$$

Step 4: Determine degrees of freedom $df = 3 - 1 - 1 = 1$ (3 genotypes, 1 allele frequency estimated from data)

Step 5: Compare to critical value Critical value at $df = 1$ is 3.84. Since $0 < 3.84$, we fail to reject the null hypothesis. The population is in HWE.

2.2 Example 6: Testing HWE with a Heterozygote Deficiency

In a population of 200 individuals, genotype counts at a locus are: AA = 110, Aa = 50, aa = 40. Test for HWE.

Step 1: Calculate observed allele frequencies Total alleles = $2 \times 200 = 400$
 $p = \frac{2 \times 110 + 50}{400} = \frac{220 + 50}{400} = \frac{270}{400} = 0.675$ $q = 1 - 0.675 = 0.325$

Step 2: Calculate expected genotype counts under HWE Expected AA: $p^2 \times 200 = (0.675)^2 \times 200 = 0.455625 \times 200 = 91.125$ Expected Aa: $2pq \times 200 = 2 \times 0.675 \times 0.325 \times 200 = 0.43875 \times 200 = 87.75$ Expected aa: $q^2 \times 200 = (0.325)^2 \times 200 = 0.105625 \times 200 = 21.125$

Step 3: Calculate chi-square

$$\begin{aligned}\chi^2 &= \frac{(110 - 91.125)^2}{91.125} + \frac{(50 - 87.75)^2}{87.75} + \frac{(40 - 21.125)^2}{21.125} \\ &= \frac{(18.875)^2}{91.125} + \frac{(-37.75)^2}{87.75} + \frac{(18.875)^2}{21.125} \\ &= \frac{356.2656}{91.125} + \frac{1425.0625}{87.75} + \frac{356.2656}{21.125} \\ &= 3.91 + 16.24 + 16.86 = 37.01\end{aligned}$$

Step 4: Determine degrees of freedom $df = 3 - 1 - 1 = 1$

Step 5: Compare to critical value Critical value at $df = 1$ is 3.84. Since $37.01 > 3.84$, we **reject** the null hypothesis. The population is NOT in HWE. The large deficiency of heterozygotes suggests possible inbreeding or population structure.

2.3 Example 7: Testing HWE with a Rare Recessive Trait

In a population of 1,000 individuals, 9 individuals show a recessive trait (aa). The observed genotype counts are: AA = 811, Aa = 180, aa = 9. Test for HWE.

Step 1: Calculate observed allele frequencies Total alleles = $2 \times 1000 = 2000$
 $p = \frac{2 \times 811 + 180}{2000} = \frac{1622 + 180}{2000} = \frac{1802}{2000} = 0.901$ $q = 1 - 0.901 = 0.099$

Step 2: Calculate expected genotype counts under HWE Expected AA: $p^2 \times 1000 = (0.901)^2 \times 1000 = 0.811801 \times 1000 = 811.80$ Expected Aa: $2pq \times 1000 = 2 \times 0.901 \times 0.099 \times 1000 = 0.178398 \times 1000 = 178.40$ Expected aa: $q^2 \times 1000 = (0.099)^2 \times 1000 = 0.009801 \times 1000 = 9.80$

Step 3: Calculate chi-square

$$\begin{aligned}\chi^2 &= \frac{(811 - 811.80)^2}{811.80} + \frac{(180 - 178.40)^2}{178.40} + \frac{(9 - 9.80)^2}{9.80} \\ &= \frac{(-0.80)^2}{811.80} + \frac{(1.60)^2}{178.40} + \frac{(-0.80)^2}{9.80} \\ &= \frac{0.64}{811.80} + \frac{2.56}{178.40} + \frac{0.64}{9.80} \\ &= 0.0008 + 0.0144 + 0.0653 \\ &= 0.0805\end{aligned}$$

Step 4: Determine degrees of freedom $df = 3 - 1 - 1 = 1$

Step 5: Compare to critical value Critical value at $df = 1$ is 3.84. Since $0.0805 < 3.84$, we **fail to reject** the null hypothesis. The population is in HWE.

3 Part 3: Degrees of Freedom in HWE Testing - A Detailed Discussion

When testing for Hardy-Weinberg equilibrium, the degrees of freedom depend on whether allele frequencies are estimated from the data or provided independently.

3.1 General Principle

For a locus with k alleles, the degrees of freedom are:

$$df = \frac{k(k+1)}{2} - 1 - m$$

where:

- $\frac{k(k+1)}{2}$ = number of possible genotypes
- -1 accounts for the constraint that total counts sum to N
- m = number of independent allele frequencies estimated from the data

For a two-allele system ($k = 2$):

- Number of genotypes = 3
- If allele frequencies are **estimated from the data**: $m = 1$ (estimating one allele frequency determines the other), so $df = 3 - 1 - 1 = 1$
- If allele frequencies are **known a priori** (not estimated from the data): $m = 0$, so $df = 3 - 1 - 0 = 2$

3.2 Why This Matters

When you know the allele frequency from an external source (e.g., from a previous study, from the parental population, or from a large reference population), you do not lose a degree of freedom for estimation. You simply test whether the observed genotype counts match the HWE expectations calculated from that known frequency.

3.3 Example 8: Allele Frequency Estimated from Data ($df = 1$)

In a population of 100 individuals, you observe: AA = 50, Aa = 30, aa = 20.

Step 1: Estimate allele frequency from the data

$$p = \frac{2 \times 50 + 30}{200} = \frac{100 + 30}{200} = \frac{130}{200} = 0.65$$

Step 2: Calculate expected counts using estimated p Expected AA = $0.65^2 \times 100 = 42.25$ Expected Aa = $2 \times 0.65 \times 0.35 \times 100 = 45.50$ Expected aa = $0.35^2 \times 100 = 12.25$

Step 3: Calculate chi-square

$$\begin{aligned}\chi^2 &= \frac{(50 - 42.25)^2}{42.25} + \frac{(30 - 45.50)^2}{45.50} + \frac{(20 - 12.25)^2}{12.25} \\ &= \frac{60.06}{42.25} + \frac{240.25}{45.50} + \frac{60.06}{12.25} \\ &= 1.42 + 5.28 + 4.90 = 11.60\end{aligned}$$

Step 4: Degrees of freedom $df = 3 - 1 - 1 = 1$

Step 5: Conclusion Critical value at $df = 1$ is 3.84. Since $11.60 > 3.84$, reject HWE.

3.4 Example 9: Allele Frequency Known from External Source ($df = 2$)

Suppose you are studying the same population, but you already know from previous large-scale studies that the true population allele frequency is $p = 0.70$. You observe: AA = 50, Aa = 30, aa = 20 (same observed counts as above).

Step 1: Use known allele frequency (not estimated from data) $p = 0.70$ (given), $q = 0.30$

Step 2: Calculate expected counts using known p Expected AA = $0.70^2 \times 100 = 49.00$ Expected Aa = $2 \times 0.70 \times 0.30 \times 100 = 42.00$ Expected aa = $0.30^2 \times 100 = 9.00$

Step 3: Calculate chi-square

$$\begin{aligned}\chi^2 &= \frac{(50 - 49)^2}{49} + \frac{(30 - 42)^2}{42} + \frac{(20 - 9)^2}{9} \\ &= \frac{1}{49} + \frac{144}{42} + \frac{121}{9} \\ &= 0.020 + 3.429 + 13.444 = 16.893\end{aligned}$$

Step 4: Degrees of freedom $df = 3 - 1 - 0 = 2$ (no allele frequencies estimated from the data)

Step 5: Conclusion Critical value at $df = 2$ is 5.99. Since $16.893 > 5.99$, reject HWE.

3.5 Example 10: Multiple Alleles with Known Frequencies

Consider an ABO blood type locus with three alleles (I^A , I^B , i). Suppose you know from historical data that the allele frequencies in a population are: $p = 0.3$ (for I^A), $q = 0.1$ (for I^B), $r = 0.6$ (for i). You sample 200 individuals and observe the following genotype counts:

Genotype	Expected Frequency	Expected Count	Observed
$I^A I^A$	$p^2 = 0.09$	18	20
$I^A I^B$	$2pq = 0.06$	12	10
$I^A i$	$2pr = 0.36$	72	70
$I^B I^B$	$q^2 = 0.01$	2	4
$I^B i$	$2qr = 0.12$	24	22
ii	$r^2 = 0.36$	72	74

Step 1: Degrees of freedom Number of genotypes = 6 Number of allele frequencies estimated from data = 0 (all given) $df = 6 - 1 - 0 = 5$

Step 2: Calculate chi-square

$$\begin{aligned}\chi^2 &= \frac{(20 - 18)^2}{18} + \frac{(10 - 12)^2}{12} + \frac{(70 - 72)^2}{72} \\ &\quad + \frac{(4 - 2)^2}{2} + \frac{(22 - 24)^2}{24} + \frac{(74 - 72)^2}{72} \\ &= \frac{4}{18} + \frac{4}{12} + \frac{4}{72} + \frac{4}{2} + \frac{4}{24} + \frac{4}{72} \\ &= 0.222 + 0.333 + 0.056 + 2.000 + 0.167 + 0.056 \\ &= 2.834\end{aligned}$$

Step 3: Conclusion Critical value at $df = 5$ is 11.07. Since $2.834 < 11.07$, fail to reject HWE.

3.6 Summary Table

Table 2: Degrees of Freedom for HWE Testing

Scenario	df (2 alleles)	df (3 alleles)
Frequencies estimated from data	1	3
Frequencies known a priori	2	5

4 Practice Problems

Mendelian Crosses

1. In a monohybrid cross between two heterozygous fruit flies ($Aa \times Aa$) for eye color (red dominant over white), a student observes 280 red-eyed and 120 white-eyed flies. Test whether these data fit the expected 3:1 Mendelian ratio.
2. In a cross between two heterozygous snapdragon plants ($Rr \times Rr$) where flower color shows incomplete dominance (RR = red, Rr = pink, rr = white), a gardener observes 35 red, 70 pink, and 35 white flowers. Test whether these data fit the expected 1:2:1 ratio.
3. In a dihybrid cross ($AaBb \times AaBb$) for seed shape and color, a researcher observes: 210 round yellow, 85 round green, 70 wrinkled yellow, 35 wrinkled green. Test whether these data fit the expected 9:3:3:1 ratio.
4. In a codominant system in a plant species, a cross between two heterozygotes ($C1C2 \times C1C2$) is expected to produce a 1:2:1 ratio. The observed offspring are: $C1C1 = 42$, $C1C2 = 96$, $C2C2 = 42$. Test whether the data fit the expected ratio.
5. In a monohybrid test cross ($Aa \times aa$) where A is dominant, the expected ratio is 1:1. A breeder observes 142 dominant phenotype and 158 recessive phenotype. Test whether these data fit the expected 1:1 ratio.

Hardy-Weinberg Equilibrium (Allele Frequencies Estimated from Data)

6. In a population of 500 individuals, the genotype counts at a codominant locus are: $AA = 180$, $AB = 240$, $BB = 80$. Test whether this population is in Hardy-Weinberg equilibrium.
7. In a population of 1,000 individuals, the following counts are observed for an autosomal locus: $AA = 490$, $Aa = 420$, $aa = 90$. Test whether this population is in HWE.
8. In a population of 200 individuals, a recessive trait is observed in 32 individuals. The observed counts are: $AA = 98$, $Aa = 70$, $aa = 32$. Test whether this population is in HWE.
9. A population of 150 individuals has the following genotype counts at a locus with two alleles: $AA = 60$, $Aa = 70$, $aa = 20$. Test whether this population is in HWE.
10. In a population of 300 individuals, the observed genotype counts are: $AA = 210$, $Aa = 80$, $aa = 10$. Test whether this population is in HWE.

Hardy-Weinberg Equilibrium (Allele Frequencies Known A Priori)

11. In a population of 150 individuals, you observe: $AA = 80$, $Aa = 60$, $aa = 10$. Previous research has established that the true allele frequency in this region is $p = 0.70$. Test whether this population is in HWE using the known allele frequency.
12. You are studying a codominant locus in a fish species. A long-term monitoring program has determined that the allele frequency of allele A is $p = 0.55$ in this lake

system. In a new sample of 400 fish, you observe: $AA = 130$, $AB = 210$, $BB = 60$. Test whether this sample is in HWE.

13. In a population of 250 plants, you observe the following genotype counts at a locus: $AA = 100$, $Aa = 125$, $aa = 25$. Calculate the chi-square value and degrees of freedom for:

- a. Testing HWE when allele frequencies are estimated from the data
- b. Testing HWE when the true allele frequency is known to be $p = 0.60$

14. At a locus with three alleles (A, B, C), you know from a reference database that the allele frequencies are: $p_A = 0.50$, $p_B = 0.30$, $p_C = 0.20$. In a sample of 100 individuals, you observe: $AA = 20$, $AB = 35$, $AC = 15$, $BB = 12$, $BC = 10$, $CC = 8$. Test whether this population is in HWE.

15. A geneticist is studying a population of butterflies. From museum specimens collected 50 years ago, the allele frequency at a wing pattern locus was determined to be $p = 0.40$. In a current sample of 200 butterflies, the following are observed: $AA = 28$, $Aa = 96$, $aa = 76$. Test whether the current population is in HWE using the historical allele frequency. What are the degrees of freedom?

5 Answer Key

Mendelian Crosses

1. Expected: 300 red, 100 white. $\chi^2 = \frac{(280-300)^2}{300} + \frac{(120-100)^2}{100} = \frac{400}{300} + \frac{400}{100} = 1.33 + 4 = 5.33$. $df = 1$, critical = 3.84. Since $5.33 > 3.84$, reject null. The data do NOT fit a 3:1 ratio.

2. Expected: 35 red, 70 pink, 35 white. $\chi^2 = \frac{(35-35)^2}{35} + \frac{(70-70)^2}{70} + \frac{(35-35)^2}{35} = 0$. $df = 2$, critical = 5.99. Fail to reject null. The data fit a 1:2:1 ratio.

3. Total = 400. Expected: 225 round yellow, 75 round green, 75 wrinkled yellow, 25 wrinkled green.

$\chi^2 = \frac{(210-225)^2}{225} + \frac{(85-75)^2}{75} + \frac{(70-75)^2}{75} + \frac{(35-25)^2}{25}$
 $\chi^2 = \frac{225}{225} + \frac{100}{75} + \frac{25}{75} + \frac{100}{25} = 1 + 1.33 + 0.33 + 4 = 6.66$. $df = 3$, critical = 7.81. Since $6.66 < 7.81$, fail to reject null. The data fit a 9:3:3:1 ratio.

4. Total = 180. Expected: 45, 90, 45. $\chi^2 = \frac{(42-45)^2}{45} + \frac{(96-90)^2}{90} + \frac{(42-45)^2}{45} = \frac{9}{45} + \frac{36}{90} + \frac{9}{45} = 0.2 + 0.4 + 0.2 = 0.8$. $df = 2$, critical = 5.99. Fail to reject null. The data fit a 1:2:1 ratio.

5. Total = 300. Expected: 150, 150. $\chi^2 = \frac{(142-150)^2}{150} + \frac{(158-150)^2}{150} = \frac{64}{150} + \frac{64}{150} = 0.427 + 0.427 = 0.854$. $df = 1$, critical = 3.84. Fail to reject null. The data fit a 1:1 ratio.

Hardy-Weinberg Equilibrium (Allele Frequencies Estimated from Data)

6. $p = \frac{2 \times 180 + 240}{1000} = \frac{360 + 240}{1000} = 0.6$, $q = 0.4$. Expected: AA = $0.36 \times 500 = 180$, AB = $0.48 \times 500 = 240$, BB = $0.16 \times 500 = 80$. $\chi^2 = 0$. $df = 1$, critical = 3.84. Fail to reject null. Population is in HWE.

7. $p = \frac{2 \times 490 + 420}{2000} = \frac{980 + 420}{2000} = \frac{1400}{2000} = 0.7$, $q = 0.3$. Expected: AA = $0.49 \times 1000 = 490$, Aa = $0.42 \times 1000 = 420$, aa = $0.09 \times 1000 = 90$. $\chi^2 = 0$. $df = 1$, critical = 3.84. Fail to reject null. Population is in HWE.

8. $p = \frac{2 \times 98 + 70}{400} = \frac{196 + 70}{400} = \frac{266}{400} = 0.665$, $q = 0.335$. Expected: AA = $(0.665)^2 \times 200 = 0.442225 \times 200 = 88.445$, Aa = $2 \times 0.665 \times 0.335 \times 200 = 0.44555 \times 200 = 89.11$, aa = $(0.335)^2 \times 200 = 0.112225 \times 200 = 22.445$.

$\chi^2 = \frac{(98-88.445)^2}{88.445} + \frac{(70-89.11)^2}{89.11} + \frac{(32-22.445)^2}{22.445}$
 $\chi^2 = \frac{91.27}{88.445} + \frac{365.13}{89.11} + \frac{91.27}{22.445} = 1.03 + 4.10 + 4.07 = 9.20$. $df = 1$, critical = 3.84. Since $9.20 > 3.84$, reject null. Population is NOT in HWE.

9. $p = \frac{2 \times 60 + 70}{300} = \frac{120 + 70}{300} = \frac{190}{300} = 0.633$, $q = 0.367$. Expected: AA = $(0.633)^2 \times 150 = 0.4007 \times 150 = 60.11$, Aa = $2 \times 0.633 \times 0.367 \times 150 = 0.4646 \times 150 = 69.69$, aa = $(0.367)^2 \times 150 = 0.1347 \times 150 = 20.20$.

$\chi^2 = \frac{(60-60.11)^2}{60.11} + \frac{(70-69.69)^2}{69.69} + \frac{(20-20.20)^2}{20.20}$
 $\chi^2 = \frac{0.0121}{60.11} + \frac{0.0961}{69.69} + \frac{0.04}{20.20} = 0.0002 + 0.0014 + 0.0020 = 0.0036$. $df = 1$, critical = 3.84. Fail to reject null. Population is in HWE.

10. $p = \frac{2 \times 210 + 80}{600} = \frac{420 + 80}{600} = \frac{500}{600} = 0.833$, $q = 0.167$. Expected: AA = $(0.833)^2 \times 300 = 0.6939 \times 300 = 208.17$, Aa = $2 \times 0.833 \times 0.167 \times 300 = 0.2782 \times 300 = 83.46$, aa = $(0.167)^2 \times 300 = 0.0279 \times 300 = 8.37$.

$\chi^2 = \frac{(210-208.17)^2}{208.17} + \frac{(80-83.46)^2}{83.46} + \frac{(10-8.37)^2}{8.37}$
 $\chi^2 = \frac{3.35}{208.17} + \frac{11.97}{83.46} + \frac{2.66}{8.37} = 0.016 + 0.143 + 0.318 = 0.477$. $df = 1$, critical = 3.84. Fail to reject null.

reject null. Population is in HWE.

Hardy-Weinberg Equilibrium (Allele Frequencies Known A Priori)

11. Known $p = 0.70$, $q = 0.30$. Expected: $AA = 0.49 \times 150 = 73.5$, $Aa = 0.42 \times 150 = 63$, $aa = 0.09 \times 150 = 13.5$.

$$\chi^2 = \frac{(80-73.5)^2}{73.5} + \frac{(60-63)^2}{63} + \frac{(10-13.5)^2}{13.5}$$

$$\chi^2 = \frac{42.25}{73.5} + \frac{9}{63} + \frac{12.25}{13.5} = 0.575 + 0.143 + 0.907 = 1.625.$$

$df = 2$ (frequencies known, not estimated). Critical = 5.99. Fail to reject HWE.

12. Known $p = 0.55$, $q = 0.45$. Expected: $AA = 0.3025 \times 400 = 121$, $AB = 2 \times 0.55 \times 0.45 \times 400 = 0.495 \times 400 = 198$, $BB = 0.2025 \times 400 = 81$.

$$\chi^2 = \frac{(130-121)^2}{121} + \frac{(210-198)^2}{198} + \frac{(60-81)^2}{81}$$

$$\chi^2 = \frac{81}{121} + \frac{144}{198} + \frac{441}{81} = 0.669 + 0.727 + 5.444 = 6.84.$$

$df = 2$, critical = 5.99. Since $6.84 > 5.99$, reject HWE.

13.

a. Estimated from data: $p = \frac{2 \times 100 + 125}{500} = \frac{200 + 125}{500} = \frac{325}{500} = 0.65$, $q = 0.35$. Expected: $AA = 0.4225 \times 250 = 105.625$, $Aa = 0.455 \times 250 = 113.75$, $aa = 0.1225 \times 250 = 30.625$.

$$\chi^2 = \frac{(100-105.625)^2}{105.625} + \frac{(125-113.75)^2}{113.75} + \frac{(25-30.625)^2}{30.625}$$

$$\chi^2 = \frac{31.64}{105.625} + \frac{126.56}{113.75} + \frac{31.64}{30.625} = 0.30 + 1.11 + 1.03 = 2.44. \quad df = 1, \text{ critical} = 3.84.$$

Fail to reject.

b. Known $p = 0.60$, $q = 0.40$. Expected: $AA = 0.36 \times 250 = 90$, $Aa = 0.48 \times 250 = 120$, $aa = 0.16 \times 250 = 40$.

$$\chi^2 = \frac{(100-90)^2}{90} + \frac{(125-120)^2}{120} + \frac{(25-40)^2}{40}$$

$$\chi^2 = \frac{100}{90} + \frac{25}{120} + \frac{225}{40} = 1.111 + 0.208 + 5.625 = 6.944. \quad df = 2, \text{ critical} = 5.99. \text{ Since } 6.944 > 5.99, \text{ reject HWE.}$$

14. Known frequencies: $p_A = 0.50$, $p_B = 0.30$, $p_C = 0.20$. Total $N = 100$.

Expected: $AA = 0.25 \times 100 = 25$, $AB = 2 \times 0.50 \times 0.30 \times 100 = 30$, $AC = 2 \times 0.50 \times 0.20 \times 100 = 20$, $BB = 0.09 \times 100 = 9$, $BC = 2 \times 0.30 \times 0.20 \times 100 = 12$, $CC = 0.04 \times 100 = 4$.

$$\chi^2 = \frac{(20-25)^2}{25} + \frac{(35-30)^2}{30} + \frac{(15-20)^2}{20} + \frac{(12-9)^2}{9} + \frac{(10-12)^2}{12} + \frac{(8-4)^2}{4}$$

$$\chi^2 = \frac{25}{25} + \frac{25}{30} + \frac{25}{20} + \frac{9}{9} + \frac{4}{12} + \frac{16}{4}$$

$$\chi^2 = 1 + 0.833 + 1.25 + 1 + 0.333 + 4 = 8.416.$$

$df = 6 - 1 - 0 = 5$, critical = 11.07. Since $8.416 < 11.07$, fail to reject HWE.

15. Known historical $p = 0.40$, $q = 0.60$. Expected: $AA = 0.16 \times 200 = 32$, $Aa = 0.48 \times 200 = 96$, $aa = 0.36 \times 200 = 72$.

$$\chi^2 = \frac{(28-32)^2}{32} + \frac{(96-96)^2}{96} + \frac{(76-72)^2}{72}$$

$$\chi^2 = \frac{16}{32} + \frac{0}{96} + \frac{16}{72} = 0.5 + 0 + 0.222 = 0.722.$$

$df = 2$ (frequencies known from external source), critical = 5.99. Fail to reject HWE.

The current population is in HWE with respect to the historical allele frequency.